Numerical computation of Hadamard finite part of some hypersingular integrals in 1, 2 and 3 dimensions

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Abstract

Data mining is a good way to find the relationship between raw data and predict the target we want which is also widely used in different field nowadays. In this project, we implement a lots of technology and method in data mining to predict the sale of an item based on its previous sale. We create a strong model to predict the sales. After evaluating this model, we conclude that this model can be used in normal life for future sale's prediction.

1 Introduction

Our project is a competition on Kaggle (Predict Future Sales). We are provided with daily historical sales data (including each products' sale date, block ,shop price and amount). And we will use it to forecast the total amount of each product sold next month. Because of the list of shops and products slightly changes every month. We need to create a robust model that can handle such situations.

2 Notations

For every physical quantity, the corresponding reference space quantity is denoted by a hat, so that, for example, if $\mathbf{x}$ is a point on a d-dimensional variety, then $\hat{\mathbf{x}}$ is the corresponding point in the reference space. If u is a function defined on the variety, then $\hat{u}$ is the corresponding function in the reference space. Furthermore, we denote by $⨎$ the Hadamard finite part of an integral.

3 Model presentation

Let's consider an finite part integral of the form

$$I = ⨎_{\hat{\tau}} f(\hat{\mathbf{y}}) \mathrm{d}\hat{\mathbf{y}}, \qquad f(\hat{\mathbf{y}}) \underset{\hat{y}\to\hat{x}}{\sim} \frac{1}{||\hat{\mathbf{y}} - \hat{\mathbf{x}}||^{d+1}}. \tag{1}$$

4 Injecting Richardson extrapolation into Guiggiani's final formula

4.1 Richardson extrapolation - general idea

For any angle θ, let's consider a finite sequence of n radial points $\rho_i(\theta)$ such that

$$0 < \rho_1(\theta) < \rho_2(\theta) < \cdots < \rho_n(\theta) < \hat{\rho}(\theta), \tag{2}$$

Recall: we can express the two first terms of the Laurent expansion of the complete polar kernel $F(\rho, \theta) = \rho K(\chi(\hat{\mathbf{x}}), \chi(\hat{\mathbf{y}}(\rho, \theta)))J(\hat{\mathbf{y}}(\rho, \theta))\hat{u}(\hat{\mathbf{y}}(\rho, \theta))$ as the following limits

$$F_{-2}(\theta) = \lim_{\rho \to 0} \rho^2 F(\rho, \theta), \qquad F_{-1}(\theta) = \lim_{\rho \to 0} \rho \left(F(\rho, \theta) - \frac{F_{-2}(\theta)}{\rho^2} \right). \tag{3}$$

such that we can write the complete polar kernel as

$$F(\rho, \theta) = \frac{F_{-2}(\theta)}{\rho^2} + \frac{F_{-1}(\theta)}{\rho} + \underset{\rho \to 1}{O}(1) \tag{4}$$

Let $c_{1,i}$ and $c_{2,i}$ be the coefficients used in the Richardson extrapolation of order n to compute the previous limits, so that we can write

$$\boxed{F_{-2}(\theta) \approx \widetilde{F}_{-2}(\theta) = \sum_{i=1}^{n} c_{2,i} \rho_i^2(\theta) F(\rho_i(\theta), \theta)}, \qquad F_{-1}(\theta) \approx \widetilde{F}_{-1}(\theta) = \sum_{i=1}^{n} c_{1,i} \rho_i(\theta) \left(F(\rho_i(\theta), \theta) - \frac{\widetilde{F}_{-2}(\theta)}{\rho_i^2(\theta)} \right). \tag{5}$$

By injecting the formula of $\widetilde{F}_{-2}(\theta)$ into the formula of $\widetilde{F}_{-1}(\theta)$, so that we can write $\widetilde{F}_{-1}(\theta)$ only as a linear combination of the complete polar kernel evaluated at the radial points $\rho_i(\theta)$, we obtain

$$\widetilde{F}_{-1}(\theta) = \sum_{i=1}^{n} c_{1,i} \rho_i(\theta) F(\rho_i(\theta), \theta) - \sum_{i=1}^{n} c_{1,i} \frac{1}{\rho_i(\theta)} \sum_{j=1}^{n} c_{2,j} \rho_j^2(\theta) F(\rho_j(\theta), \theta).$$

By letting $P_n(\theta) = \sum_{k=1}^{n} \frac{1}{\rho_k(\theta)}$ be a constant that is a polynomial in $\cos\theta$ and $\sin\theta$, we can write the previous formula as

$$\boxed{\widetilde{F}_{-1}(\theta) = \sum_{i=1}^{n} F(\rho_i(\theta), \theta) \rho_i(\theta) \Big(c_{1,i} - c_{2,i} P_n(\theta) \rho_i(\theta) \Big)} \tag{6}$$

4.2 Lagrange basis method

For this section and the following, we will no longer use θ in the notation, it will be implicit. Let's denote the regular function $G(\rho) = \rho^2 F(\rho)$, so that $G(0) = F_{-2}$ and $G'(0) = F_{-1}$. Let $\ell_i(\rho)$ be the Lagrange basis of degree $n-1$ associated to the radial points ρ_i, defined as

$$\ell_i(\rho) = \prod_{j \neq i} \frac{\rho - \rho_j}{\rho_i - \rho_j}, \qquad i = 1, \ldots, n. \tag{7}$$

The lagrange interpolant of G and its derivative are given by

$$\Pi_{n-1} G(\rho) = \sum_{i=1}^{n} G(\rho_i) \ell_i(\rho), \qquad (\Pi_{n-1} G)'(\rho) = \sum_{i=1}^{n} G(\rho_i) \ell_i'(\rho). \tag{8}$$

So that the numerical Laurent's coefficients, denoted $\widetilde{F}_{-2}$ and $\widetilde{F}_{-1}$, are simply the evaluation of the interpolant and its derivative at the origin $\rho = 0$:

$$\widetilde{F}_{-2} = \Pi_{n-1}G(0) = \sum_{i=1}^{n} G(\rho_i)\ell_i(0), \qquad \widetilde{F}_{-1} = (\Pi_{n-1}G)'(0) = \sum_{i=1}^{n} G(\rho_i)\ell_i'(0). \tag{9}$$

with $\ell_i(0) = \prod_{j\neq i} \frac{\rho_j}{\rho_j - \rho_i}$ and $\ell_i'(0) = \ell_i(0)\left(\frac{1}{\rho_i} - P_n\right)$, so that we can finally write the Laurent's coefficients as

$$\widetilde{F}_{-2} = \sum_{i=1}^{n} \ell_i(0)\rho_i^2 F(\rho_i), \qquad \widetilde{F}_{-1} = \sum_{i=1}^{n} \ell_i(0)\rho_i\left(1 - P_n\rho_i\right)F(\rho_i). \tag{10}$$

This is a particular case of the Richardson extrapolation, with the two families of coefficients $c_{1,i}$ and $c_{2,i}$ collapsing into a single family of coefficients $\ell_i(0)$:

$$c_{1,i} = c_{2,i} = \ell_i(0) = c_i = \prod_{j\neq i} \frac{\rho_j}{\rho_j - \rho_i}.$$

Finally, the numerical Laurent's coefficients can be written as

$$\widetilde{F}_{-2}(\theta) = \sum_{i=1}^{n} A_i(\theta)F(\rho_i(\theta), \theta), \qquad \widetilde{F}_{-1}(\theta) = \sum_{i=1}^{n} B_i(\theta)F(\rho_i(\theta), \theta). \tag{11}$$

With $A_i(\theta) = c_i(\theta)\rho_i^2(\theta)$ and $B_i(\theta) = c_i(\theta)\rho_i(\theta)\left(1 - P_n(\theta)\rho_i(\theta)\right)$.

Each of these formula simply consist in the numerical evaluation of the polar kernel at the radial points $\rho_i(\theta)$, multiplied by some weights.

4.2.1 Properties of $A_i(\theta)$ and $B_i(\theta)$

$$\sum_{i=1}^{n} A_i = \begin{cases} \rho_1, & n = 1 \\ -\rho_1\rho_2, & n = 2 \\ 0, & n \geq 3 \end{cases}, \qquad \sum_{i=1}^{n} B_i = \begin{cases} 0, & n = 1 \\ \rho_1 + \rho_2, & n = 2 \\ 0, & n \geq 3 \end{cases} \tag{12}$$

These three cases illustrates the following definition :

$$\sum_{i=1}^{n} A_i = (\Pi_{n-1}[\rho^2])(0), \qquad \sum_{i=1}^{n} B_i = (\Pi_{n-1}[\rho^2])'(0) \tag{13}$$

4.3 Final quadrature rule

Let's then denote $I(\theta)$ the integral of the complete polar kernel over the radial direction, defined as:

$$I(\theta) = \int_0^{\hat{\rho}(\theta)} \left\{ F(\rho, \theta) - \frac{F_{-1}(\theta)}{\rho} - \frac{F_{-2}(\theta)}{\rho^2} \right\} \mathrm{d}\rho + F_{-1}(\theta)\ln\left|\frac{\hat{\rho}(\theta)}{\beta(\theta)}\right| - F_{-2}(\theta)\left\{\frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)}\right\} \tag{14}$$

A final quadrature rule for it is the following:

$$I(\theta) \approx \widetilde{I}(\theta) = \sum_{i=1}^{n} \widetilde{w}_i(\theta)F(\rho_i(\theta), \theta) \tag{15}$$

With the new weights $\widetilde{w}_i(\theta)$ defined as

$$\widetilde{w}_i(\theta) = w_i(\theta) - A_i(\theta)\left\{W_{-2,n}(\theta) + \frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)}\right\} + B_i(\theta)\left\{\ln\left|\frac{\hat{\rho}(\theta)}{\beta(\theta)}\right| - W_{-1,n}(\theta)\right\}$$

and the auxiliary constants :

- $w_i(\theta) = \frac{\hat{\rho}(\theta)}{2} w_i^{\text{Ref}}$ are the weights of the reference quadrature over the segment $[-1, 1]$.
- $W_{-1,n}(\theta) = \sum_{i=1}^{n} \frac{w_i(\theta)}{\rho_i(\theta)} = \sum_{i=1}^{n} \frac{w_i^{\text{Ref}}}{1+x_i^{\text{Ref}}}$.
- $W_{-2,n}(\theta) = \sum_{i=1}^{n} \frac{w_i(\theta)}{\rho_i^2(\theta)} = \frac{2}{\hat{\rho}(\theta)} \sum_{i=1}^{n} \frac{w_i^{\text{Ref}}}{\left(1+x_i^{\text{Ref}}\right)^2}$.

Remark: in the case of Gauss-Legendre quadrature, the constants $W_{-1,n}$ and $W_{-2,n}$ can be computed analytically as:

$$W_{-1,n} = 2H_n, \qquad W_{-2,n}(\theta) = \frac{2}{\hat{\rho}(\theta)} n(n+1), \qquad P_n(\theta) = \frac{n(n+1)}{\hat{\rho}(\theta)} \tag{16}$$

With $H_n = \sum_{i=1}^{n} \frac{1}{i}$ the n-th harmonic number. These last two formulas are, for the moment, a conjecture. (Elaborate to prove it?)

4.4 Properties of the new weights $\widetilde{w}_i(\theta)$

$$\text{For } n \geq 3, \quad \sum_{i=1}^{n} \widetilde{w}_i(\theta) = \hat{\rho}(\theta) \tag{17}$$

This is easy to understand: either look at the definition of $\widetilde{w}_i(\theta)$ and use the properties of $A_i(\theta)$ and $B_i(\theta)$, or integrate the constant function $F(\rho, \theta) = 1$ between 0 and $\hat{\rho}(\theta)$, which is equivalent. In a more general way, if the integrand F is not singular (i.e. $F \sim \mathcal{O}(1)$), then the quadrature rule is the standard quadrature as if reference quadrature was used to integrate the function F between 0 and $\hat{\rho}(\theta)$. If the integrand is singular as $1/\rho$, the customed quadrature just performs a Cauchy principal value. The true ingredients are the values of the Laurent coefficients F_{-1} and F_{-2}, which are computed numerically; they can be zero (up to the algorithm precision) or not, depending on the considered singularity.

4.5 Convergence and error study

4.5.1 General idea

To begin our reasoning, assume that we have computed the Laurent's coefficients up to a relative error ϵ_{-k}, $k \in \{1, 2\}$:

$$\widetilde{F}_{-k} = F_{-k}(1 + \epsilon_{-k}), \qquad k \in \{1, 2\} \tag{18}$$

Now let's denote

$$R(\rho, \theta) = F(\rho, \theta) - \frac{F_{-1}(\theta)}{\rho} - \frac{F_{-2}(\theta)}{\rho^2} \tag{19}$$

$$\widetilde{R}(\rho, \theta) = F(\rho, \theta) - \frac{\widetilde{F}_{-1}(\theta)}{\rho} - \frac{\widetilde{F}_{-2}(\theta)}{\rho^2} \tag{20}$$

$$\tag{21}$$

$$I = \int_0^{2\pi} \int_0^{\hat{\rho}(\theta)} R(\rho, \theta) \mathrm{d}\rho \mathrm{d}\theta, \qquad \widetilde{I} = \int_0^{2\pi} \int_0^{\hat{\rho}(\theta)} \widetilde{R}(\rho, \theta) \mathrm{d}\rho \mathrm{d}\theta \tag{22}$$

On one hand, after the theory of the Bernstein's ellipses, if we assume that R and $\widetilde{R}$ are analytic, we can make a error majoration regarding the Gauss-Legendre quadrature of I and $\widetilde{I}$:

$$|I - Q_n[R]| \leq \frac{64}{15} \frac{M}{(1-\rho_B^{-2})\rho_B^{2n}} \tag{23}$$

$$\left|\widetilde{I} - Q_n[\widetilde{R}]\right| \leq \frac{64}{15} \frac{\widetilde{M}}{(1-\widetilde{\rho}_B^{-2})\widetilde{\rho}_B^{2n}} \tag{24}$$

On the other hand, we can substract $\widetilde{I}$ to I to get:

$$|I - \widetilde{I}| \leq C(\hat{x}) \sum_{k=1}^{2} \frac{\epsilon_{-k}}{\rho_{\min}^k} ||F_{-k}||_\infty \tag{25}$$

where $C(\hat{x}) = \int_0^{2\pi} \hat{\rho}(\theta) \mathrm{d}\theta$.

Thus,

$$|I - Q_n[\widetilde{R}]| = |I - \widetilde{I} + \widetilde{I} - Q_n[\widetilde{R}]| \quad \leq |I - \widetilde{I}| + |\widetilde{I} - Q_n[\widetilde{R}]| \leq C(\hat{x}) \sum_{k=1}^{2} \frac{\epsilon_{-k}}{\rho_{\min}^k} ||F_{-k}||_\infty + \frac{64}{15} \frac{\widetilde{M}}{(1-\widetilde{\rho}_B^{-2})\widetilde{\rho}_B^{2n}} \tag{26}$$

4.5.2 Richardson extrapolation

Lemma 1:

We can show that the following majorations hold for any sequence $\rho_i(\theta)$ of n radial points:

$$|\widetilde{F}_{-2} - F_{-2}| \leq \frac{||\partial_\rho^n G||_\infty}{n!} \prod_{i=1}^{n} \rho_i, \qquad |\widetilde{F}_{-1} - F_{-1}| \leq \left\{ \frac{\left\|\partial_\rho^{n+1} G\right\|_\infty}{(n+1)!} + P_n \frac{\left\|\partial_\rho^n G\right\|_\infty}{n!} \right\} \prod_{i=1}^{n} \rho_i \tag{27}$$

Which shows that the speed of convergence is conditionned by the product of the radial points $\prod_{i=1}^n \rho_i(\theta)$. In the case of a geometric sequence $\rho_i(\theta) = \hat{\rho}(\theta) q^{i-1}$, we have $\prod_{i=1}^n \rho_i(\theta) = \hat{\rho}^n(\theta) q^{\frac{n(n-1)}{2}}$, which is a very fast convergence. In the case of a sequence of Gauss-Legendre quadrature points, we have $\prod_{i=1}^n \rho_i(\theta) = \hat{\rho}^n(\theta) \frac{(n!)^2}{(2n)!}$, which is a slower convergence (see Figure 1).

That's why geometric sequences are still preferred over Gauss-Legendre quadrature points for pure Richardson extrapolation.

Proof of Lemma 1: Let $e(\rho) = G(\rho) - \Pi_{n-1} G(\rho)$ be the interpolation error of G by its Lagrange interpolant $\Pi_{n-1} G$. We have the following formula for the interpolation error, which is quite general on polynomial interpolation:

$$e(\rho) = G[\rho_1, \cdots, \rho_n, \rho] \prod_{i=1}^{n} (\rho - \rho_i) \tag{28}$$

Where $G[\rho_1, \cdots, \rho_n, \rho]$ is the divided difference of order n of G at the points $\rho_1, \cdots, \rho_n, \rho$. By using the mean value theorem for divided differences, we can write

$$G[\rho_1, \cdots, \rho_n, \rho] = \frac{G^{(n)}(\xi)}{n!} \tag{29}$$

For some $\xi \in [0, \hat{\rho}]$. Thus, we can write

$$\left|\widetilde{F}_{-2} - F_{-2}\right| = |e(0)| \leq \frac{||\partial_\rho^n G||_\infty}{n!} \prod_{i=1}^{n} \rho_i \tag{30}$$

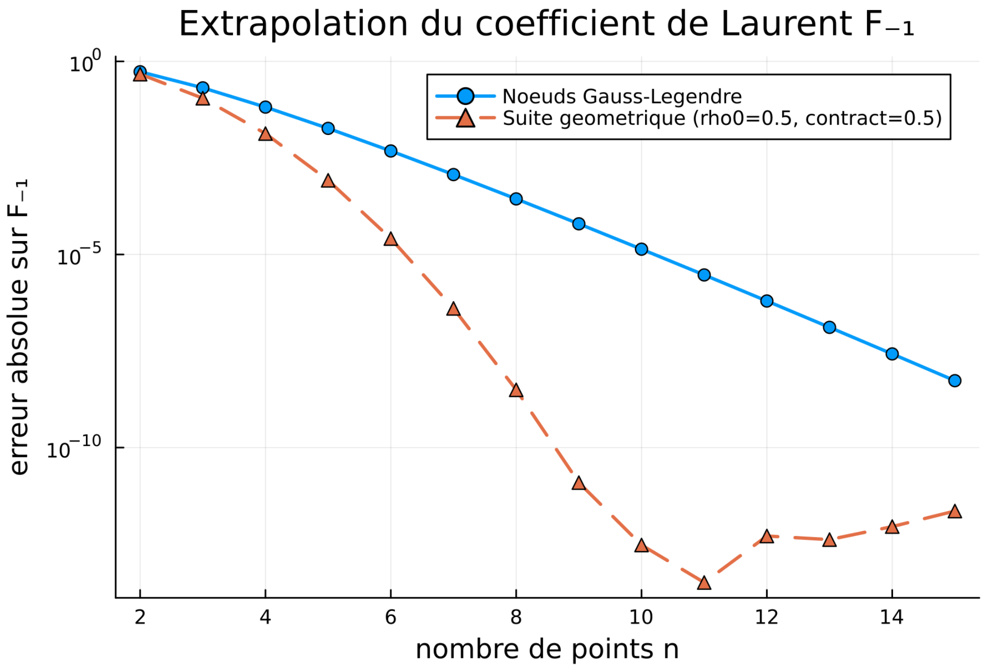

Figure 1: Convergence comparison between geometric and Gauss-Legendre sequences for the function $\rho \to 1/(1+\rho)$

Then, taking the derivative of the interpolation error, we have

$$
\begin{aligned}
e'(\rho) &= \partial_\rho \left\{ G[\rho_1, \cdots, \rho_n, \rho] \prod_{i=1}^{n} (\rho - \rho_i) \right\} \\
&= \partial_\rho \left\{ G[\rho_1, \cdots, \rho_n, \rho] \right\} \prod_{i=1}^{n} (\rho - \rho_i) + G[\rho_1, \cdots, \rho_n, \rho] \partial_\rho \left\{ \prod_{i=1}^{n} (\rho - \rho_i) \right\} \\
&= G[\rho_1, \cdots, \rho_n, \rho, \rho] \prod_{i=1}^{n} (\rho - \rho_i) + G[\rho_1, \cdots, \rho_n, \rho] P_n \prod_{i=1}^{n} (\rho - \rho_i) \\
\left| \widetilde{F}_{-1} - F_{-1} \right| = |e'(0)| &\leq \left\{ \frac{\left\| \partial_\rho^{n+1} G \right\|_\infty}{(n+1)!} + P_n \frac{\left\| \partial_\rho^{n} G \right\|_\infty}{n!} \right\} \prod_{i=1}^{n} \rho_i
\end{aligned}
$$

using again the mean value theorem for $G[\rho_1, \cdots, \rho_n, \rho]$ and $G[\rho_1, \cdots, \rho_n, \rho, \rho]$.

4.5.3 Gauss-Legendre quadrature

To delve deeper into these general ideas, let's consider :

$$
\widetilde{I}_{\epsilon,h,n}(\theta) = Q_n[\widetilde{R}_{\epsilon,h}](\theta) = \sum_{i=1}^{n} w_i(\theta) \left\{ F_h(\rho_i(\theta), \theta) - \frac{\widetilde{F}_{-1,\epsilon,h}(\theta)}{\rho_i(\theta)} - \frac{\widetilde{F}_{-2,\epsilon,h}(\theta)}{\rho_i^2(\theta)} \right\} \tag{31}
$$

$$
= \sum_{i=1}^{n} \left\{ F_h(\rho_i(\theta), \theta) - \frac{F_{-1,h}(\theta)}{\rho_i(\theta)} - \frac{F_{-2,h}(\theta)}{\rho_i^2(\theta)} \right\} w_i(\theta) - \sum_{k=1}^{2} F_{-k,h}(\theta) \epsilon_{-k} \sum_{i=1}^{n} \frac{w_i(\theta)}{\rho_i^k(\theta)} \tag{32}
$$

$$
= Q_n[R_h](\theta) - \sum_{k=1}^{2} F_{-k,h}(\theta) \epsilon_{-k} W_{-k,n}(\theta) \tag{33}
$$

Where h is the mesh size. Thanks to standard results on the Gauss-Legendre quadrature for any function in $\mathcal{C}^{2n}([0,\hat{\rho}(\theta)])$, we have the following error majoration (deleting the θ dependency for clarity):

$$Q_n[R_h] = \int_0^{\hat{\rho}} R_h(\rho)\mathrm{d}\rho + \frac{2^{2n+1}(n!)^4}{(2n+1)[(2n)!]^3} R_h^{(2n)}(\rho_0) \tag{34}$$

for some $\rho_0 \in [0,\hat{\rho}]$. Our next goal is to show that the error term $R_h^{(2n)}(\rho_0)$ converges with the mesh size h and the number of quadrature points n. By using $G_h(\rho) = \rho^2 F_h(\rho) = \mathcal{O}_{\rho\to 0}(1)$, we can straightforwardly express R_h in terms of G, and bound its derivatives, assuming, as Guiggiani, that F_h is analytic in a neighborhood of the origin:

$$R_h(\rho) = \sum_{k=0}^{+\infty} F_k \rho^k, \qquad \Longrightarrow \qquad R_h^{(2n)}(\rho) = \sum_{k=2n}^{+\infty} F_k \frac{k!}{(k-2n)!}\rho^{k-2n} \tag{35}$$

$$\tag{36}$$

with $F_k = \dfrac{G_h^{(k+2)}(0)}{(k+2)!} \le \dfrac{1}{(k+2)!}||G_h^{(k+2)}||_{\infty,[0,\hat{\rho}]}$ and $\rho^{k-2n} \le \hat{\rho}^{k-2n}$, so that we can write

$$|R_h^{(2n)}(\rho)| \le \sum_{k=2n}^{+\infty} k!\, \frac{||G_h^{(k+2)}||_{\infty,[0,\hat{\rho}]}}{(k+2)!\,(k-2n)!}\hat{\rho}^{k-2n} \tag{37}$$

BOUNDING OF G AND ITS DERIVATIVES: let us defined the following functions:

$$\begin{array}{cccc} \phi_h & : & \hat{\tau}\subset\mathbb{R}^{d-1} & \to & \mathbb{R} \\ & & \hat{x} & \mapsto & \det M_h(\hat{x}) \end{array}, \qquad \begin{array}{cccc} \psi_{\hat{x}} & : & \mathbb{R}\times[0,2\pi[& \to & \mathbb{R}^{d-1} \\ & & (\rho,\theta) & \mapsto & \hat{x}+\rho\mathbf{u}(\theta) \end{array}$$

$$\begin{array}{ccccc} \chi_h & : & \hat{\tau}\subset\mathbb{R}^{d-1} & \to & \tau\subset\mathbb{R}^d \\ & & \hat{x} & \mapsto & x \end{array}$$

where χ is the mapping from the reference space to the physical element, $\psi_{\hat{x}}$ is the mapping from polar coordinates to the reference space, and M_h is the metric tensor of the mapping χ_h, defined as:

$$M_h(\hat{x}) = D\chi_h(\hat{x})^T D\chi_h(\hat{x}), \qquad (M_h)_{ab} = \sum_{k=1}^{d-1} \frac{\partial \chi_{h,k}}{\partial \hat{x}_a}\frac{\partial \chi_{h,k}}{\partial \hat{x}_b}, \quad (a,b)\in\{1,\dots,d-1\}^2 = \{1,2\}^2 \tag{38}$$

Finally, the vector $\mathbf{u}(\theta)$ is the unit trigonometric vector defined as $\mathbf{u}(\theta) = (\cos\theta, \sin\theta)$ for our case $d=3$.

Lemma 1: using the Faà di Bruno formula for functions of the form $h(x) = f[g_1(x),\cdots,g_m(x)]$ (see Constantine, 1996), one can write, for any function $f:\mathbb{R}^{d-1}\to\mathbb{R}$,

$$\partial_\rho^k (f\circ\psi_{\hat{x}})(\rho,\theta) = \sum_{i=0}^{k}\binom{k}{i}\cos^i(\theta)\sin^{k-i}(\theta)\frac{\partial^k f}{\partial \hat{x}_1^i \partial \hat{x}_2^{k-i}}(\psi_{\hat{x}}(\rho,\theta)) \tag{39}$$

We can then expand G as

$$G(\rho,\theta) = H_h(\rho,\theta) J_h \circ \psi_{\hat{x}}(\rho,\theta) \tag{40}$$

where

$$H_h(\rho,\theta) = \rho^3 K\Big(x=\chi_h(\hat{x}), y = \chi_h\circ\psi_{\hat{x}}(\rho,\theta)\Big)\hat{u}\circ\psi_{\hat{x}}(\rho,\theta), \quad J_h(\hat{x}) = \sqrt{\phi_h(\hat{x})} = \sqrt{\det M_h(\hat{x})} \tag{41}$$

Lemma 2: We have the following useful factorization:

$$\boldsymbol{r} = \rho \boldsymbol{A}_{\hat{x}}(\rho,\theta), \quad \boldsymbol{A}_{\hat{x}}(\rho,\theta) = \sum_{m\geq 1} \frac{\rho^{m-1}}{m!} D^m \chi_h(\hat{x}) :_m \left(\boldsymbol{u}(\theta)^{m\otimes}\right) \tag{42}$$

Homogeneous kernels. When the kernel is homogeneous, i.e

$$\exists \hat{K}, K(x,y) = \frac{1}{r^s}\hat{K}(\hat{\mathbf{r}}, n_x, n_y), \quad \forall n_x, n_y \in \mathbb{S}^d, z \to \hat{K}(z, n_x, n_y) \text{ is of class } \mathcal{C}^\infty(\mathbb{R}^d)$$

$$\text{i.e } \forall \alpha \in \mathbb{N}^d, \quad \exists C_K(|\alpha|) > 0, \quad \left|\frac{\partial^{|\alpha|}\hat{K}}{\partial z^\alpha}(z, n_x, n_y)\right| \leq C_K(|\alpha|), \quad \forall z \in \mathbb{R}^d \tag{43}$$

Then we have

$$\rho^s K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho,\theta)) = \left(\frac{\rho}{r}\right)^s \hat{K}\left(\hat{\mathbf{A}}_{\hat{x}}(\rho,\theta), n\circ\chi_h(\hat{x}), n\circ\chi_h\circ\psi_{\hat{x}}(\rho,\theta)\right) \tag{44}$$

$$= \frac{1}{\|\boldsymbol{A}_{\hat{x}}(\rho,\theta)\|^s}\hat{K}\left(\hat{\mathbf{A}}_{\hat{x}}(\rho,\theta), n\circ\chi_h(\hat{x}), n\circ\chi_h\circ\psi_{\hat{x}}(\rho,\theta)\right) \tag{45}$$

$$= A_{\hat{x}}(\rho,\theta)^{-s}\hat{K}\left(\hat{\mathbf{A}}_{\hat{x}}(\rho,\theta), n\circ\chi_h(\hat{x}), n\circ\chi_h\circ\psi_{\hat{x}}(\rho,\theta)\right) \tag{46}$$

Where s is the order of singularity of the function $\rho \to K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho,\theta))$ at the origin, such that $K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho,\theta)) = \mathcal{O}_{\rho\to 0}(\rho^{-s})$. The idea was originally used for hypersingular 3D kernels, where $s = 3$ (see Guiggiani, 1992), but we write the computations with general order s because the idea remains the same.

We will denote $\hat{K}\left(\hat{\mathbf{A}}_{\hat{x}}(\rho,\theta), n\circ\chi_h(\hat{x}), n\circ\chi_h\circ\psi_{\hat{x}}(\rho,\theta)\right)$ as $\hat{K}_{\hat{x}}(\rho,\theta)$ for clarity, keeping in mind that this is the kernel evaluated in polar coordinates in parameter space centered at point $\hat{x}$.

As the kernel K is regularized by the term ρ^3 in H_h, we can ensure that $H_h = \mathcal{O}_{\rho\to 0}(1)$. Applying the Leipniz formula, we have

$$\partial_\rho^m G(\rho,\theta) = \sum_{k=0}^{m}\binom{m}{k}\partial_\rho^k H_h(\rho,\theta)\partial_\rho^{m-k}(J_h\circ\psi_{\hat{x}})(\rho,\theta) \tag{47}$$

With:

$$\partial_\rho^k H_h(\rho,\theta) \leq u_k \sum_{j=0}^{k}\binom{k}{j}\partial_\rho^j\left(A_{\hat{x}}(\rho,\theta)^{-s}\hat{K}_{\hat{x}}(\rho,\theta)\right)$$

$$\partial_\rho^{m-k}(J_h\circ\psi_{\hat{x}})(\rho,\theta) \leq \sum_{j=0}^{m-k}\binom{m-k}{j}\left|\frac{\partial^{m-k}J_h}{\partial\hat{x}_1^i\hat{x}_2^{m-k-i}}(\psi_{\hat{x}}(\rho,\theta))\right|$$

Where $u_k = \max_{|\boldsymbol{\alpha}|\leq k}\|\partial^{\boldsymbol{\alpha}}\hat{u}\|_{L^\infty(\hat{\tau})}$ is a constant that depends only on the geometry of the element. To bound the $(m-k)$-th order polar derivatives of J_h, we used the lemma 39.

BOUNDING OF THE CARTHESIAN DERIVATIVES OF J_h: there's no need to expand an explicit formula for the derivatives of J_h in terms parametrization χ_h, because we can use a useful hypothesis on the mesh quality. Indeed, to ensure non-degeneracy of the elements, we can assume that

$$\exists C_Q > 0, \quad \forall h > 0, \quad \forall m > 0, \quad \forall \hat{x} \in \hat{\tau}, \quad \|D^m\chi_h(\hat{x})\| \leq C_Q h^m \tag{48}$$

Where $D^m\chi(\hat{x})$ is the tensor of order m containing all the m-th order derivatives of χ_h at $\hat{x}$, defined as

$$(D^m \chi_h(\hat{x}))_{i,\alpha_1,\cdots\alpha_m} = \frac{\partial^m \chi_{h,i}}{\partial \hat{x}_{\alpha_1} \cdots \partial \hat{x}_{\alpha_m}}(\hat{x}), \qquad i \in \{1,2,3\}, \quad (\alpha_1,\cdots,\alpha_m) \in \{1,2\}^m \tag{49}$$

In particular, for $\alpha = 1$, we have

$$\|M_h(\hat{x})\| = \left\|D\chi_h(\hat{x})^T D\chi_h(\hat{x})\right\| \leq C_Q^2 h^2. \tag{50}$$

Condition 55 can be rewritten as $\boxed{r \geq c_Q h \rho}$, which represents the bi-Lipschitz property of the parametrization χ_h. Furthermore, the Hadamard's inequality (HI) states that

$$|\phi_h(\hat{x})| = |\det(M_h(\hat{x}))| \overset{\text{(HI)}}{\leq} \prod_{i=1}^{d-1} \|M_h(\hat{x})_{i,:}\| \leq \|M_h(\hat{x})\|^{d-1} \leq C_Q^{2(d-1)} h^{2(d-1)} \tag{51}$$

Remarks : theses inequalities holds perfectly if we consider the ∞-norm, but all the norms are equivalent in finite dimension, so we can use any norm; that leads to different majoration constants but the idea remains the same. This ensure that we can define the following function:

$$\begin{array}{rccl} \mu & : & \hat{\tau} \subset \mathbb{R}^{d-1} & \to & \mathbb{R} \\ & & \hat{x} & \mapsto & \sqrt{h^{-2(d-1)} \det M_h(\hat{x})} = h^{-(d-1)} J_h(\hat{x}) \leq C_Q^{d-1} \end{array}$$

By the non-degeneracy hypothesis, we have $\mu = \mathcal{O}_{h\to 0}(1)$ and all its derivatives are also bounded. So that we can write $J_h(\hat{x}) = h^{d-1}\mu(\hat{x})$, and

$$\partial_\rho^{m-k}(J_h \circ \psi_{\hat{x}})(\rho,\theta) \leq \mu_m h^{d-1} \sum_{j=0}^{m-k} \binom{m-k}{j} = \lambda_m h^{d-1} 2^{m-k} \tag{52}$$

Where $\mu_m = \max\limits_{|\boldsymbol{\alpha}|\leq m} \|\partial^{\boldsymbol{\alpha}} \mu\|_{L^\infty(\hat{\tau})}$. We can then obtain:

$$\partial_\rho^m G(\rho,\theta) \leq h^{d-1} \mu_m u_m \sum_{k=0}^{m} \sum_{j=0}^{k} 2^{m-k} \binom{m}{k} \binom{k}{j} \partial_\rho^j \left(A_{\hat{x}}(\rho,\theta)^{-s} \hat{K}_{\hat{x}}(\rho,\theta) \right) \tag{53}$$

By reordering the sums on the domain $0 \leq j \leq k \leq m$, and using the binomial identity $\binom{m}{k}\binom{k}{j} = \binom{m}{j}\binom{m-j}{k-j}$, we finally get:

$$\partial_\rho^m G(\rho,\theta) \leq h^{d-1} \mu_m u_m \sum_{j=0}^{m} \binom{m}{j} 3^{m-j} \partial_\rho^j \left(A_{\hat{x}}(\rho,\theta)^{-s} \hat{K}_{\hat{x}}(\rho,\theta) \right) \tag{54}$$

Then, we recall the shape-regularity condition which the upper bound equivalent of the non-degeneracy condition 55:

$$\exists\, c_Q > 0, \quad \forall h > 0, \quad \forall (\hat{x},\hat{y}) \in \hat{\tau}^2, \quad \|\chi_h(\hat{x}) - \chi_h(\hat{y})\| \geq c_Q\, h\, \|\hat{x} - \hat{y}\|. \tag{55}$$

Lemma 3:

$$\forall \boldsymbol{\lambda} \in \mathbb{N}^d, \forall \boldsymbol{x} \in \mathbb{R}^d \setminus \{\boldsymbol{O}\}, \qquad \left|\partial^{\boldsymbol{\lambda}} \|\boldsymbol{x}\|_2\right| \leq \boldsymbol{\lambda}! \left(1 + \frac{1}{\sqrt{2}}\right) \sqrt{2}^{|\boldsymbol{\lambda}|} \|\boldsymbol{x}\|_2^{1-|\boldsymbol{\lambda}|} \tag{56}$$

Where $\boldsymbol{\lambda}! = \prod_{i=1}^d \lambda_i!$ and $|\boldsymbol{\lambda}| = \sum_{i=1}^d \lambda_i$.
Remark: The majoration constant $C_{\boldsymbol{\lambda}} = \boldsymbol{\lambda}! \left(1 + \frac{1}{\sqrt{2}}\right) \sqrt{2}^{|\boldsymbol{\lambda}|}$ is not optimal, but it is better than the one that would be obtained after using the Faà di Bruno formula.

Proof of Lemma 3. Let $\boldsymbol{x} \in \mathbb{R}^d \setminus \{\boldsymbol{O}\}$. First, the function $\|\cdot\|$ is analytic at $\boldsymbol{x}$, so by extension it admits an holomorphic continuation in a neighborhood of $\boldsymbol{x}$ in $\mathbb{C}^d \setminus \Sigma$, where Σ is the singular complex variety defined as $\Sigma = \{\boldsymbol{z} \in \mathbb{C}^d, \sum_{i=1}^d z_i^2 = 0\}$. By the Cauchy's estimate, if we have an open ball $B(\boldsymbol{x}, R) \subset \mathbb{C}^d \setminus \Sigma$, we have the following majoration for any multi-index $\boldsymbol{\lambda} \in \mathbb{N}^d$:

$$\left|\partial^{\boldsymbol{\lambda}} \|\boldsymbol{x}\|_2\right| \leq \frac{\boldsymbol{\lambda}!}{R^{|\boldsymbol{\lambda}|}} \sup_{\boldsymbol{z} \in B(\boldsymbol{x},R)} \|\boldsymbol{z}\|_2 \tag{57}$$

If we want to make this inequality more explicit, we should find such a radius R that depends only on $\boldsymbol{x}$. Let's compute the distance from $\boldsymbol{x}$ to the singular variety Σ. Let $\boldsymbol{z} = \boldsymbol{y} + i\boldsymbol{c} \in \Sigma$.

$$\begin{aligned}
\boldsymbol{z} \in \Sigma &\iff \boldsymbol{z} \cdot \boldsymbol{z} = 0 \\
&\iff (\boldsymbol{y} + i\boldsymbol{c}) \cdot (\boldsymbol{y} + i\boldsymbol{c}) = 0 \\
&\iff |\boldsymbol{y}|^2 - |\boldsymbol{c}|^2 = 2i\, \boldsymbol{y} \cdot \boldsymbol{c} \\
&\iff \begin{cases} |\boldsymbol{y}|^2 - |\boldsymbol{c}|^2 = 0, \\ \boldsymbol{y} \cdot \boldsymbol{c} = 0. \end{cases}
\end{aligned}$$

Furthermore, we have

$$\begin{aligned}
|\boldsymbol{x} - \boldsymbol{z}|^2 &= |\boldsymbol{x} - \boldsymbol{y} - i\boldsymbol{c}|^2 \\
&= |\boldsymbol{x} - \boldsymbol{y}|^2 + |\boldsymbol{c}|^2
\end{aligned}$$

So $\operatorname{dist}(\boldsymbol{x}, \Sigma)^2 = \inf\limits_{\substack{\boldsymbol{y},\boldsymbol{c} \in \mathbb{R}^d \\ |\boldsymbol{y}|=|\boldsymbol{c}|,\ \boldsymbol{y} \perp \boldsymbol{c}}} |\boldsymbol{x} - \boldsymbol{y}|^2 + |\boldsymbol{c}|^2 = \inf\limits_{\substack{\boldsymbol{y},\boldsymbol{c} \in \mathbb{R}^d \\ \boldsymbol{y} \perp \boldsymbol{c}}} |\boldsymbol{x} - \boldsymbol{y}|^2 + |\boldsymbol{y}|^2$. For $d \geq 2$, for every $\boldsymbol{y} \in \mathbb{R}^d$ it always exists a $\boldsymbol{c} \in \mathbb{R}^d$ such that $\boldsymbol{y} \perp \boldsymbol{c}$. So we can write

$$\operatorname{dist}(\boldsymbol{x}, \Sigma)^2 = \inf_{\boldsymbol{y} \in \mathbb{R}^d} |\boldsymbol{x} - \boldsymbol{y}|^2 + |\boldsymbol{y}|^2$$

If we let $f(\boldsymbol{y}) = |\boldsymbol{x} - \boldsymbol{y}|^2 + |\boldsymbol{y}|^2$, we have $\nabla f(\boldsymbol{y}) = 4\boldsymbol{y} - 2\boldsymbol{x}$. The minimum is reached for $\nabla f(\boldsymbol{y}) = 0$, i.e. $\boldsymbol{y} = \frac{\boldsymbol{x}}{2}$. Thus, we have

$$\boxed{\operatorname{dist}(\boldsymbol{x}, \Sigma) = \frac{|\boldsymbol{x}|}{\sqrt{2}}}$$

This formula gives us how close we are from the "danger zone" in the complex space. Thus, we let $R = \theta \|\boldsymbol{x}\|_2$ with $0 < \theta < 1/\sqrt{2}$ as a relaxing parameter, as well as $C(\theta) = \frac{\boldsymbol{\lambda}!}{R^{|\boldsymbol{\lambda}|}} \sup\limits_{\boldsymbol{z} \in B(\boldsymbol{x},R)} \|\boldsymbol{z}\|_2$. For all $\boldsymbol{z} \in B(\boldsymbol{x}, R)$, we have $\|\boldsymbol{z}\|_2 \leq \|\boldsymbol{x}\|_2 + R = (1+\theta)\|\boldsymbol{x}\|_2$[1], so that we can write

$$\sup_{\boldsymbol{z} \in B(\boldsymbol{x},R)} \|\boldsymbol{z}\|_2 \leq (1+\theta)\|\boldsymbol{x}\|_2$$

and also $C(\theta) \leq \boldsymbol{\lambda}! \dfrac{1+\theta}{\theta^{|\boldsymbol{\lambda}|}} \|\boldsymbol{x}\|_2^{1-|\boldsymbol{\lambda}|} = \boldsymbol{\lambda}!\, g(\theta) \|\boldsymbol{x}\|_2^{1-|\boldsymbol{\lambda}|}$. The function g is decreasing on $]0, 1/\sqrt{2}[$, so the best majorant that we can take is the minimum majorant reached at the maximum value of θ, i.e. $\theta = 1/\sqrt{2}$, which gives us $g(1/\sqrt{2}) = (1 + 1/\sqrt{2})\sqrt{2}^{|\boldsymbol{\lambda}|}$. Thus, we have the desired majoration 56.

Corollary 1: Let $\boldsymbol{f} : \mathbb{R} \to \mathbb{R}^d$ be a function of class $\mathcal{C}^{|\boldsymbol{\lambda}|}$, with $\boldsymbol{\lambda} \in \mathbb{N}^d$. Then, the following bound holds, which is a conequence of the three following statement:

[1]Here, we perform a non optimal majoration.

- if $\boldsymbol{v}, \boldsymbol{u} \in \mathbb{R}^d$, then vectorial exponentitation can be simply bounded by $|\boldsymbol{u}^{\boldsymbol{v}}| \leq \|\boldsymbol{u}\|_2^{\|\boldsymbol{v}\|_1} = \|\boldsymbol{u}\|_2^{|\boldsymbol{v}|}$
- the Faà di Bruno formula for the composition of functions $h = g \circ f$ with $g : \mathbb{R}^d \to \mathbb{R}$ and $f : \mathbb{R} \to \mathbb{R}^d$:
- the previous lemma 56

$$|\partial^n \|\boldsymbol{f}(\rho)\|_2| \leq \sum_{1 \leq |\boldsymbol{\lambda}| \leq n} \boldsymbol{\lambda}!\, C_{\boldsymbol{\lambda}} \|f(\rho)\|_2^{1-|\boldsymbol{\lambda}|} \sum_{p(n,\boldsymbol{\lambda})} n! \prod_{j=1}^{n} \frac{\|\partial^j \boldsymbol{f}(\rho)\|_2^{|\boldsymbol{k}_j|}}{\boldsymbol{k}_j!\, j!^{|\boldsymbol{k}_j|}} \tag{58}$$

Lemma 4: We have the following useful lower and upper bounds: $\forall \boldsymbol{k}_1, \ldots, \boldsymbol{k}_n \in p(n, \boldsymbol{\lambda})$

$$\prod_{i=1}^{n} (i+1)!^{|\boldsymbol{k}_i|} \geq \Gamma\left(\frac{n}{|\boldsymbol{\lambda}|} + 2\right)^{|\boldsymbol{\lambda}|} \tag{59}$$

$$\prod_{i=1}^{n} (i+1)^{|\boldsymbol{k}_i|} \leq \left(\frac{n}{|\boldsymbol{\lambda}|} + 1\right)^{|\boldsymbol{\lambda}|} \tag{60}$$

Where Γ is the Euler's Gamma function: $\Gamma(x) = \displaystyle\int_0^{+\infty} t^{x-1} e^{-t} \mathrm{d}t$.

Lemma 5: We can achieve the following upper bound for the derivatives of scalar $A_{\hat{x}}(\rho, \theta)$:

$$\left|\partial_\rho^n A_{\hat{x}}(\rho, \theta)\right| \leq c_Q h^n \sum_{1 \leq |\boldsymbol{\lambda}| \leq n} C_{\boldsymbol{\lambda}} |p(n, \boldsymbol{\lambda})| n!\, e^{|\boldsymbol{\lambda}| \hat{\rho}(\theta) h} \Gamma\left(\frac{n}{|\boldsymbol{\lambda}|} + 2\right)^{-|\boldsymbol{\lambda}|}$$

Where $C_{\boldsymbol{\lambda}} = \boldsymbol{\lambda}! \left(1 + \dfrac{1}{\sqrt{2}}\right) \sqrt{2}^{|\boldsymbol{\lambda}|}$.

After some rearrangements, and assuming that it exists a constant $C_{n,|\boldsymbol{\lambda}|}$ depending only on n and $|\boldsymbol{\lambda}|$ such that $|p(n, \boldsymbol{\lambda})| \leq C_{n,|\boldsymbol{\lambda}|}$, we can write, as $\boldsymbol{\lambda}! \leq |\boldsymbol{\lambda}|!$, the following:

$$\left|\partial_\rho^n A_{\hat{x}}(\rho, \theta)\right| \leq h^{n+1} \mathcal{C}_n$$

with $\mathcal{C}_n = \mathcal{C}_n(h, \theta, \hat{x}, \chi_h) = c_Q \displaystyle\sum_{k=1}^{n} \binom{n+k-1}{k} C_k C_{n,k} \left(\frac{e^{h\hat{\rho}(\theta)}}{\Gamma\left(\frac{k}{n} + 2\right)}\right)^k$ and $C_k = k! \left(1 + \frac{1}{\sqrt{2}}\right) \sqrt{2}^k$.

Also, we have $\mathcal{C}_n = \mathcal{O}_{h \to 0}(1)$ and $\mathcal{C}_n \neq \mathcal{O}_{h \to 0}(h)$. We made the change of variable $k = |\boldsymbol{\lambda}|$, and $\binom{n+k-1}{k} = \# \{\boldsymbol{\lambda} \in \mathbb{N}^n, |\boldsymbol{\lambda}| = k\}$.

STUDY OF THE DERIVATIVES OF THE REGULARIZED KERNEL: Let's differentiate $A_{\hat{x}}(\rho, \theta)^{-s} \hat{K}_{\hat{x}}(\rho, \theta)$. Using both the Leibniz and the Faà di Bruno formulas, we can write

$$\left(\frac{f}{g^s}\right)^{(n)} = \sum_{i=0}^{n} \binom{n}{i} f^{(n-i)} \sum_{k=0}^{i} (-1)^k g^{-s-k} s^{(k)} \sum_{p(i,k)} \frac{i!}{\prod_{j=1}^{i} k_j!} \prod_{j=1}^{i} \left(\frac{\partial^j g}{j!}\right)^{k_j}$$

With, by convention,

$$\sum_{p(i,0)} \frac{i!}{\prod_{j=1}^{i} k_j!} \prod_{j=1}^{i} \left(\frac{\partial^j g}{j!}\right)^{k_j} = \delta_{i,0},$$

so that we can retrieve, for $n = 0$, the original function $\dfrac{f}{g^s}$, and where $s^{(k)} = s(s+1)\cdots(s+k-1) = \dfrac{(s+k-1)!}{(s-1)!}$ is the rising factorial.

Before diving into these whole summations and product, let's establish a few straightforward bounds. Let's first write the polar equivalence (it obviously exists) of the bounding condition 43:

$$\exists C_{n-i} > 0, \qquad \partial_\rho^{n-i} \hat{K}_{\hat{x}}(\rho,\theta) \leq \hat{C}_{n-i} = C_{n-i}(\theta, \hat{x}, \chi_h) \tag{61}$$

Then, by using the mesh regularity condition 55, we can write

$$A_{\hat{x}}(\rho,\theta)^{-s-k} = (r/\rho)^{-s-k} \leq (c_Q h)^{-s-k} \tag{62}$$

Now, our next goal is to bound the term $P_{p(i,k)} = \frac{1}{\prod_{j=1}^{i} k_j!} \prod_{j=1}^{i} \left(\frac{\partial_\rho^j A_{\hat{x}}(\rho,\theta)}{j!} \right)^{k_j}$.

$$P_{p(i,k)} = \prod_{j=1}^{i} \frac{\left(\partial_\rho^j A_{\hat{x}}(\rho,\theta)\right)^{k_j}}{j!^{k_j}\, k_j!} \leq \prod_{j=1}^{i} \frac{\left(h^{j+1} \mathcal{C}_j\right)^{k_j}}{j!^{k_j}\, k_j!} \tag{63}$$

On one hand, $\prod_{j=1}^{i} \left(h^{j+1}\mathcal{C}_j\right)^{k_j} = h^{k+i} \prod_{j=1}^{i} \mathcal{C}_j^{k_j}$, by definition of the set $p(i,k)$. On the other hand, by lemma 59,

$$\prod_{j=1}^{i} k_j!\, j!^{k_j} \geq \prod_{j=1}^{i} j!^{k_j} = \prod_{j=1}^{i} \left((j+1)!\right)^{k_j} / \prod_{j=1}^{i} (j+1)^{k_j} \overset{59}{\geq} \Gamma\left(\frac{i}{k}+2\right)^{k} \left(\frac{i}{k}+1\right)^{-k}$$

Thus,

$$\sum_{p(i,k)} P_{p(i,k)} \leq h^{k+i} \mathcal{C}_{i,k} \left(\frac{\frac{i}{k}+1}{\Gamma\left(\frac{i}{k}+2\right)} \right)^{k}, \qquad \mathcal{C}_{i,k} = \sum_{p(i,k)} \prod_{j=1}^{i} \mathcal{C}_j^{k_j}, \qquad k \overset{\text{recall}}{=} \sum_{j=1}^{i} k_j \tag{64}$$

Finally,

$$\partial_\rho^j \left(A_{\hat{x}}(\rho,\theta)^{-s} \hat{K}_{\hat{x}}(\rho,\theta) \right) \leq \left(\frac{1}{h c_Q} \right)^{s} \sum_{i=0}^{j} (j)_i \hat{C}_{j-i} h^i \sum_{k=1}^{i} s^{(k)} \mathcal{C}_{i,k} \left(\frac{\frac{i}{k}+1}{\Gamma\left(\frac{i}{k}+2\right)} \right)^{k} \tag{65}$$

With $(k)_i = \frac{k!}{(k-i)!}$ the falling factorial ; which gives, at the end,

$$\partial_\rho^m G(\rho,\theta) \leq h^{d-s-1} \frac{\mu_m u_m}{c_Q^s} \sum_{0 \leq k \leq i \leq j \leq m} h^i \binom{m}{j} 3^{m-j} s^{(k)} (j)_i \hat{C}_{j-i} \mathcal{C}_{i,k} \left(\frac{i/k+1}{\Gamma(i/k+2)} \right)^{k} \tag{66}$$

$$= \underset{h \to 0}{O}\left(h^{d-s-1}\right) \tag{67}$$

again with the convention

$$\mathcal{C}_{i,0} = \delta_{i,0}$$